

SMART INDIA HACKATHON 2026 | PROBLEM STATEMENT ID: 26087
MINISTRY OF COOPERATION | NATIONAL COUNCIL FOR COOPERATIVE TRAINING

SahakarSetu: AI & LMS-Enabled Cooperative Capacity Building, ERP & Employment Ecosystem

Comprehensive Technical Architecture & Solution Engineering Report
Hardware Category | Theme: Smart Education | Mode: Software + Hardware

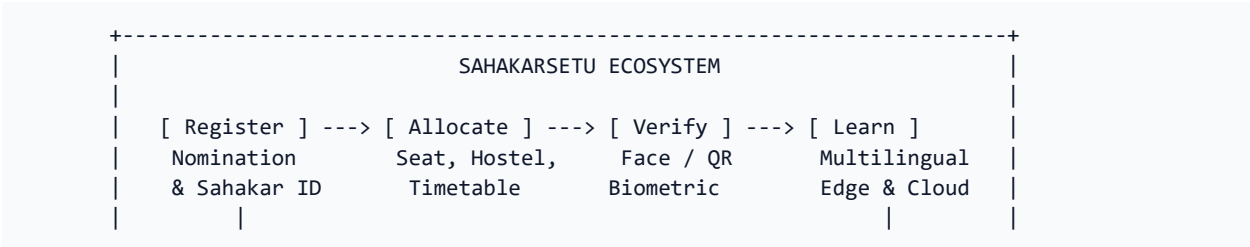
Project Deployment & Prototype Links

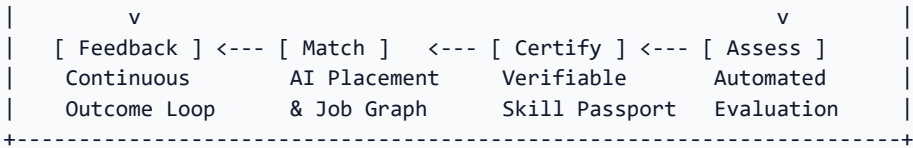
1. Executive Summary & System Vision

SahakarSetu is an scalable digital ecosystem designed to modernize and unify the cooperative education, administration, and employment ecosystem across India. SahakarSetu is a proposed digital ecosystem designed to support the cooperative training ecosystem under NCCT and the Ministry of Cooperation. the platform integrates institutional enterprise resource planning (ERP), learning management systems (LMS), computer-vision attendance, offline-first edge computing, verifiable cryptographic skill credentialing, and AI-driven employment linkage into a continuous, data-driven lifecycle.

NCCT operates through its apex institution—Vaikunth Mehta National Institute of Cooperative Management (VAMNICOM), Pune—alongside 5 Regional Institutes of Cooperative Management (RICMs) and 14 Institutes of Cooperative Management (ICMs). In total, these 20 institutions conduct more than 3,700 programmes annually, training over 2.27 lakh participants comprising Primary Agricultural Credit Societies (PACS) secretaries, dairy cooperative staff, women Self-Help Groups (SHGs), cooperative bank officers, and rural youth.

Historically, this vast educational apparatus has operated through manual or siloed systems, leading to paper attendance registers, duplicate trainee records, zero centralized learning analytics, lack of verified skill portability, and minimal direct linkage to formal employment or entrepreneurship.





SahakarSetu bridges the critical rural infrastructure gap through a hybrid **Software + Hardware** architecture. The core innovation is the **Sahakar Edge Box**—a low-cost, ruggedized, low-power edge computer deployed at each training institute, PACS center, and rural cooperative hub. The Edge Box runs an embedded micro-cloud with localized LMS media caching, biometric face/QR attendance validation, automated quiz grading, and a store-and-forward SQLite journal. Trainees in remote, low-bandwidth areas connect locally via Wi-Fi without active internet. When connectivity is restored, the Edge Box cryptographically synchronizes with the central Sahakar Cloud, enabling zero-downtime operations and national-level training monitoring.

2. Problem Statement & Domain Context

2.1 Background

The Ministry of Cooperation has spearheaded nationwide digitalization, including computerization of 63,000+ functional PACS, establishment of Model Bye-laws, creation of the National Cooperative Database (cataloging over 8.44 lakh cooperatives and 30+ crore members), and introduction of the National Cooperation Policy 2025.

NCCT serves as the intellectual backbone for this cooperative revolution. However, field operations face significant operational bottlenecks:

- 1. Administrative Fragmentation:** Trainee registration, institution approvals, hostel room allocations, timetable generation, and logistics tracking are maintained in disconnected spreadsheets or physical logbooks.
- 2. Connectivity Deficits:** Rural PACS and remote ICM centers suffer from intermittent or non-existent broadband, rendering purely cloud-dependent software unusable in remote rural classrooms.
- 3. Attendance Integrity:** Paper attendance and single-point biometrics are vulnerable to proxy check-ins, lack audit trails, and fail during network outages.
- 4. Credential Fraud & Lack of Portability:** Paper certificates are difficult for prospective cooperative banks and agri-employers to authenticate, lacking verifiable skill breakdown.
- 5. Employment Disconnect:** While cooperative societies urgently require certified talent in PACS accounting, Tally ERP, inventory management, and cold chain logistics, trained rural youth have no direct visibility into vacancies.

2.2 Problem Statement & Scope

To design, develop, and implement an integrated, scalable, web-based digital ecosystem for cooperative training institutions that combines:

- Centralized ERP for institutional administration (nominations, timetables, hostels, logistics).

- Offline-first Learning Management System (LMS) with interactive multilingual content.
- Hardware-enabled Edge computing for offline training continuity.
- AI-based computer vision (face recognition with liveness) and cryptographic dynamic QR attendance.
- Cryptographically verifiable Digital Skill Passports.
- AI-driven skill-to-job matching engine and RAG-based career guidance.
- Centralized analytics dashboard for national training impact and policy planning.

3. Requirement-to-Solution Mapping

The table below demonstrates direct, full-scope alignment with the technical requirements of Problem Statement 26087:

#	Requirement (PS 26087)	SahakarSetu Engineering Solution	Technical Implementation
1	Online Registration & Nomination	Web-based nomination portal supporting self-registration, society sponsorship, and institute bulk approvals.	React 19 PWA, Django REST APIs, automated eligibility screening.
2	Participant, Institution & Trainee Profiles	Centralized national registry issuing a unique, immutable lifetime digital Sahakar ID.	PostgreSQL central ledger, cryptographic format `SAH-YYYY-INST-XXXXXX`.
3	Interactive Multilingual E-Learning	SCORM-compliant LMS with video lectures, interactive modules, and regional voice narration.	Proposed Moodle LMS Core integration + Proposed Government BHASHINI AI speech/translation API.
4	Digital Attendance (Face / QR)	Dual-mode attendance engine: on-device ArcFace recognition with liveness check, plus rotating cryptographic QR fallback.	Proposed InsightFace / MiniFASNet embeddings + time-based HMAC dynamic QR codes.
5	Timetable, Hostel & Logistics Management	Campus ERP module managing classroom scheduling, faculty allocation, hostel room beds, and training kit dispatches.	Relational allocation engine with conflict resolution and GPS-tagged logistics.
6	LMS Integration, Assessment & Certification	Automated module quizzes, proctored summative exams, rubric-based grading, and instant certificate issuing.	LMS evaluation pipeline enforcing customizable cutoff score benchmarks.
7	Skill Certification Repository	Digital Skill Passport with	Ed25519 digitally signed

	& Verification	tamper-evident cryptographic signature and instant public QR code lookup.	credentials with public-key verification, public verification endpoint.
8	Career Counseling Chatbot	AI knowledge assistant providing guidance on cooperative schemes, bye-laws, and career paths.	Proposed Retrieval-Augmented Generation (RAG) using LangChain / pgvector over NCCT docs.
9	Employer & Recruiter Dashboard	Dedicated recruiter portal for posting vacancies, filtering certified candidates, and reviewing Skill Passports.	Cooperative recruitment hub with automated candidate pipeline management.
10	Mobile-Friendly & Offline-Accessible Platform	Mobile-responsive PWA backed by the Sahakar Edge Box for zero-connectivity classrooms.	React PWA + Raspberry Pi 5 local Wi-Fi micro-cloud with store-and-forward sync.
11	Centralized Database & Monitoring Analytics	Real-time national intelligence center tracking mobilization, certification, sync rates, and placements.	Recharts analytics engine fed by central PostgreSQL event warehouse.

4. End-to-End System Architecture

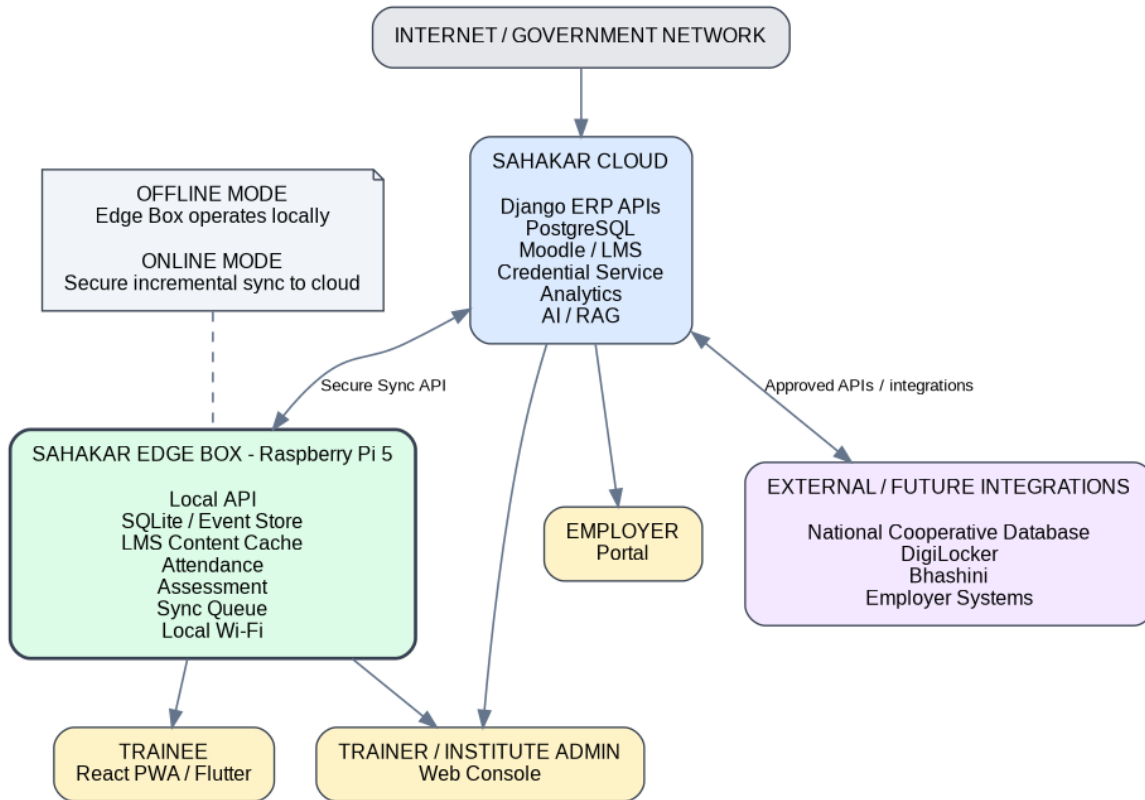
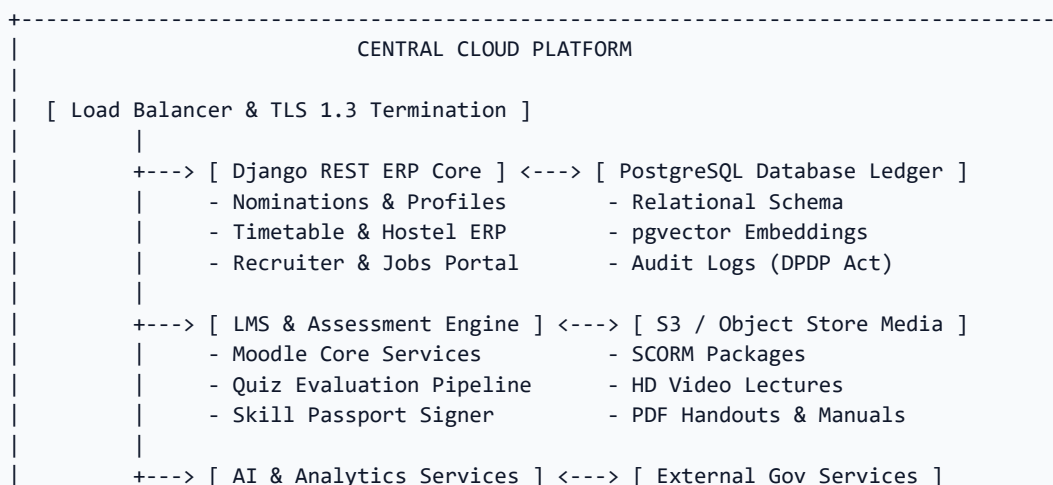
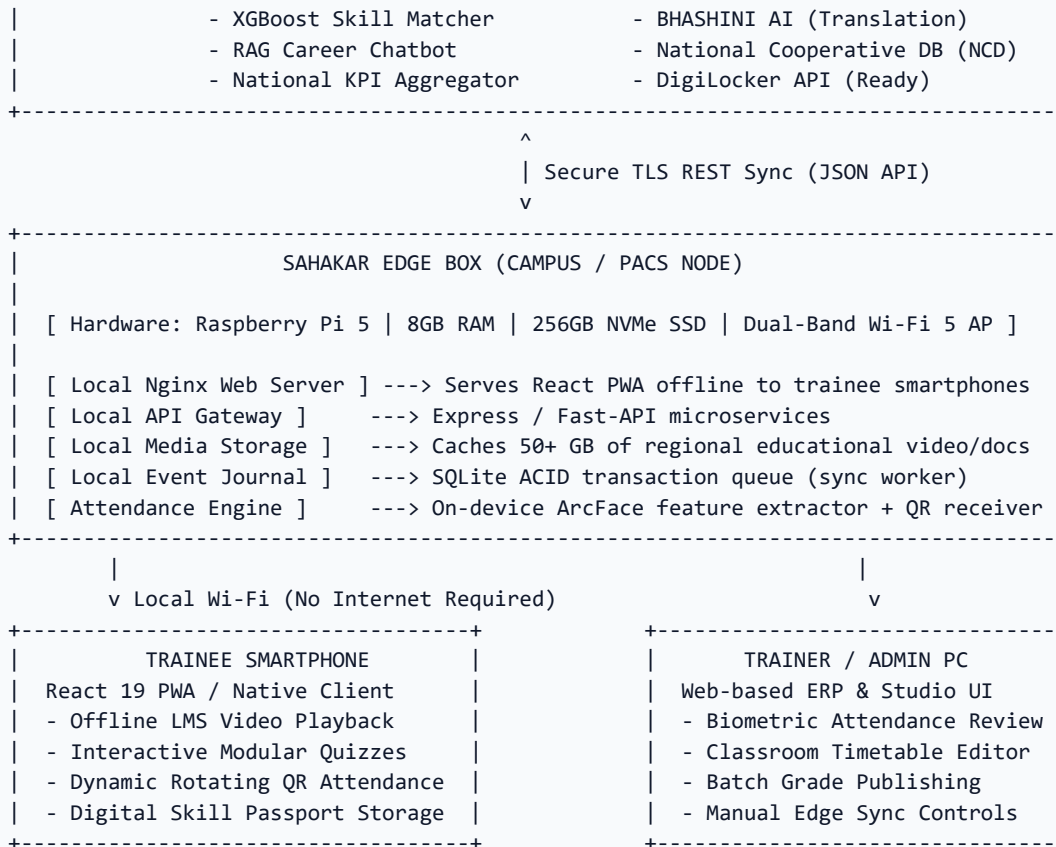


Figure 1: SahakarSetu End-to-End Hybrid System Architecture (Cloud + Edge Box)

4.1 Architectural Blueprint

SahakarSetu is built on a resilient, multi-tiered micro-cloud architecture designed for high throughput, offline durability, and resilient, offline-capable architecture that allows local training operations to continue during network/cloud outages:





4.2 Multi-Tier Logical Stack

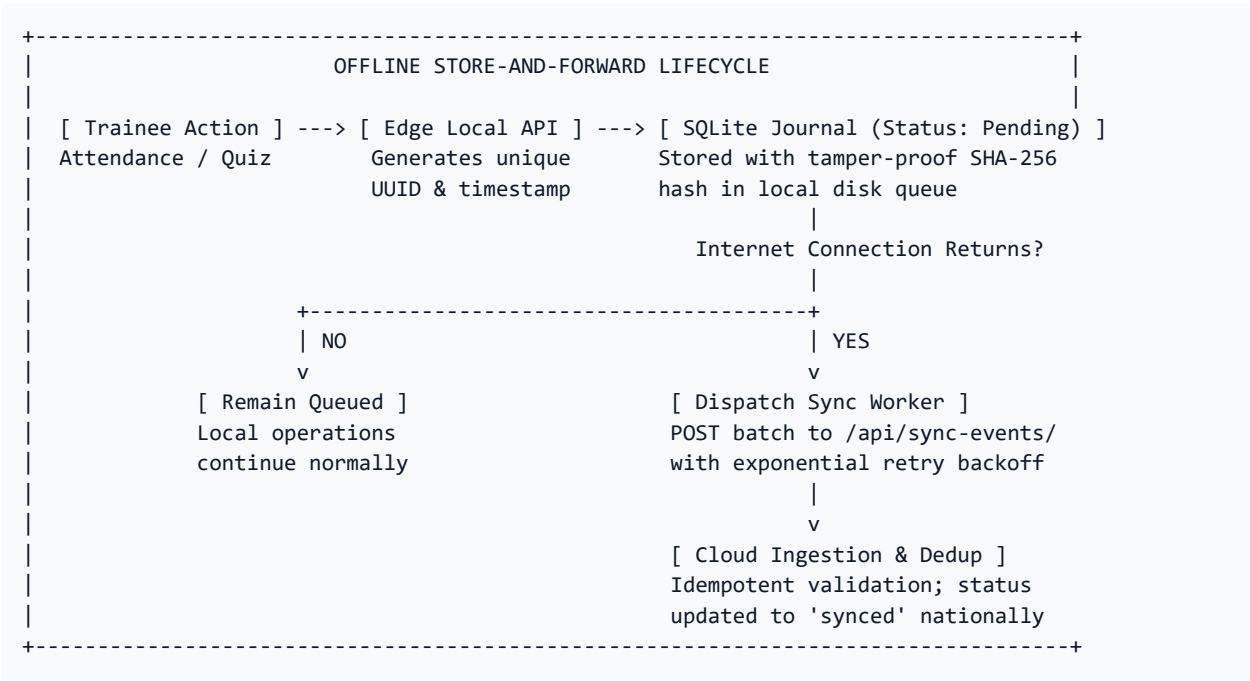
- Edge Presentation Tier:** React 19 Progressive Web Application (PWA) compiled to static assets, featuring Service Workers for client-side offline storage (`idb`/IndexedDB), responsive Tailwind CSS UI, and touch-optimized components.
- Edge Compute Tier:** Raspberry Pi 5 node running Debian/Ubuntu Linux, providing a local captive Wi-Fi portal (SSID: `SahakarSetu-EdgeBox`), local Nginx web server, and asynchronous background store-and-forward workers.
- Cloud Application Tier:** Python 3.11 / Django 5 REST Framework modular monolith running in containerized environments (Gunicorn + ASGI), exposing 40+ standardized REST endpoints.
- Cloud Persistence Tier:** PostgreSQL 16 relational database with ACID compliance, row-level security, JSONB fields for dynamic offline payloads, and `pgvector` extension for semantic embeddings.
- AI Inference Pipeline:**
 - **Speech & Dialect:** Government BHASHINI REST APIs for speech-to-text, text-to-speech, and Indic machine translation.
 - **Vision & Biometrics:** InsightFace / ArcFace lightweight neural network (MobileNetV2 backbone) producing 512-dimensional Euclidean face vectors with MiniFASNet liveness verification.

- Skill Matching: XGBoost Learning-to-Rank (`rank:ndcg`) algorithm operating over ESCO v1.2.1 and O*NET taxonomies.

5. Sahakar Edge Box - Hardware Core Engineering

5.1 Purpose & Operating Principle

In India's cooperative training topography, rural institutes and remote village PACS often face network outages lasting hours or days. The **Sahakar Edge Box** functions as an intelligent decentralized gateway. Instead of denying access when the internet drops, training operations continue seamlessly in local autonomous mode.



5.2 Hardware Specification & Bill of Materials (BOM)

The Edge Box is engineered with commercial off-the-shelf (COTS) industrial components to guarantee low cost, modular repairability, and ease of mass assembly:

Item	Component	Specification	Function	Indicative Cost
1	Single Board Computer	Raspberry Pi 5 (8GB LPDDR4X)	Quad-core ARM Cortex-A76 @ 2.4GHz; 8GB RAM	Core edge compute node
2	Active Cooling & Enclosure	Aluminum Heatsink + PWM Fan + ABS Case	Ruggedized industrial casing with heat dissipation	Hardware protection

3	High-Speed Storage	PCIe NVMe M.2 SSD (256 GB) via M.2 HAT	256 GB PCIe NVMe SSD via M.2 HAT high-speed local storage for offline LMS media and SQLite journal.	Offline video cache & SQLite
4	Networking	Dual-Band Wi-Fi 5 (802.11ac) + Gigabit Ethernet	2.4/5.0 GHz Access Point (up to 60 concurrent clients)	Local trainee wireless portal
5	Real-Time Clock (RTC)	I2C DS3231 RTC Module with battery backup	High precision TCXO RTC with coin cell	Offline timestamp integrity
6	Power Resilience	5V/5A USB-C PD Adapter + Mini 18650 UPS	10,000 mAh battery buffer (4-6 hours backup)	Uninterrupted power operation
7	Biometric / QR Camera Kiosk	5MP Wide-Angle Camera + Rotating Stand	1080p @ 30fps with autofocus and LED ring	Attendance face/QR terminal
Total	Complete Sahakar Edge Box Unit	Ruggedized Plug-and-Play Assembly	Autonomous Rural Training Node	~₹19,450

5.3 Offline Synchronization & Collision Resolution

1. **Idempotency Guarantee:** Every action occurring offline is assigned a cryptographically unique identifier at origin:

$$\text{Event ID} = \text{UUIDv4} \parallel \text{Timestamp} \parallel \text{DeviceID}$$

2. **Ordered Event Processing:** Events are written to an append-only SQLite transaction table with statuses: `pending`, `syncing`, `synced`, `conflict`.

3. **Dynamic Reconnection Worker:** A background daemon constantly monitors network health by pinging `https://sahakarsetu-mcdr.onrender.com/health/`. Upon ping acknowledgement, the worker dispatches batches of 50 events using HTTP POST to `/api/sync-events/`.

4. **Collision Handling:** If an attendance record or enrollment action was processed independently on both cloud and edge, the central server applies the **Latest Valid Cryptographic Signature (LVCS)** rule, preserving data integrity without human intervention.

6. Unified Trainee Profile & Digital Identity (Sahakar ID)

6.1 Identity Structure

Every participant entering the NCCT training ecosystem receives a lifetime unique digital identity called the **Sahakar ID**. This replaces ad-hoc local register serials with a standardized, verifiable identifier across all 20 institutes.

\text{Sahakar ID Format: } \mathbf{SAH - [YYYY] - [INST_CODE] - [SERIAL_NUMBER]}

Example: `SAH-2026-VAM01-001847` (Trainee registered in 2026 at VAMNICOM Pune, Serial #1847).

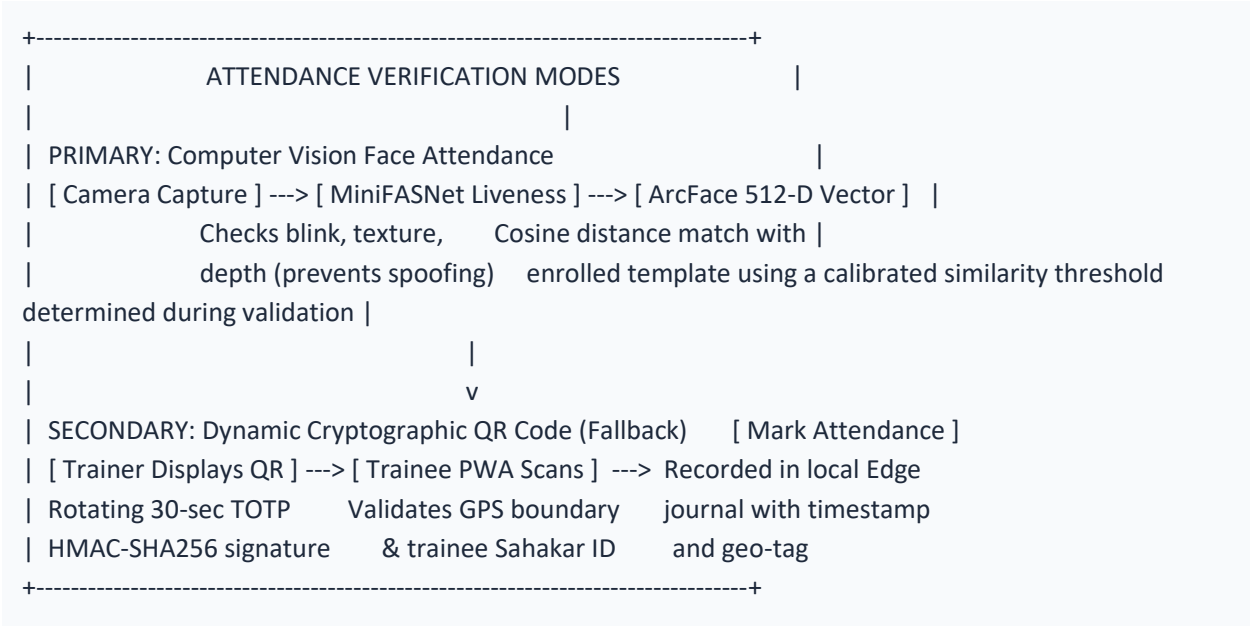
6.2 Data Model & Linkage

The Sahakar ID acts as the primary key linking:

- Demographic & cooperative federation membership (e.g., Baramati Taluka Milk Union).
- Multi-institute historical course enrollments (e.g., PACS Accounting 2025, Dairy ERP 2026).
- Time-stamped biometric attendance records.
- Completed module quizzes and final exam scores.
- Verifiable digital skill credentials and Skill Passport badges.
- Active cooperative employer job applications and placement confirmations.

7. Dual-Mode Attendance System (Computer Vision + Dynamic QR)

To eliminate attendance fraud, proxy check-ins, and manual paperwork, SahakarSetu implements a dual-mode verification pipeline tailored for institutional campuses and rural outreach centers:



7.1 Mode A: ArcFace Computer Vision with Liveness Detection

- **Liveness Screening:** Prior to feature extraction, frames are analyzed by **MiniFASNet**, an ultra-lightweight convolutional network that detects printed paper attacks, tablet replays, and silicone masks by computing micro-surface reflectance and eye-blink frequency.
- **Biometric Vector Extraction:** Faces passing liveness are processed by an **ArcFace (Additive Angular Margin Loss)** neural network, converting the face image into a compact 512-dimensional floating-point embedding:

$$L_{\text{ArcFace}} = -\log \frac{e^{s(\cos(\theta_{y_i} + m))}}{e^{s(\cos(\theta_{y_i} + m))} + \sum_{j \neq y_i} e^{s(\cos \theta_j)}}$$

- **Privacy Preservation (DPDP Compliance):** Raw photographs are discarded immediately from memory after embedding generation. Only the non-reversible 512-dimensional numerical vector is stored, encrypted using AES-256.

7.2 Mode B: Rotating Cryptographic QR Code

For high-density classrooms (60+ trainees entering simultaneously in a 5-minute window), the instructor's terminal renders a rotating QR code updated every 30 seconds.

- The QR payload embeds: `InstituteID | CourseID | SessionID | UTC\_Timestamp | HMAC-SHA256\_Signature`.
- The trainee's PWA scans the QR code; the device cross-verifies that the phone's GPS coordinates fall within the geofenced classroom radius (≤ 50 meters). Attendance is stamped instantly.

8. E-Learning & LMS Core Architecture

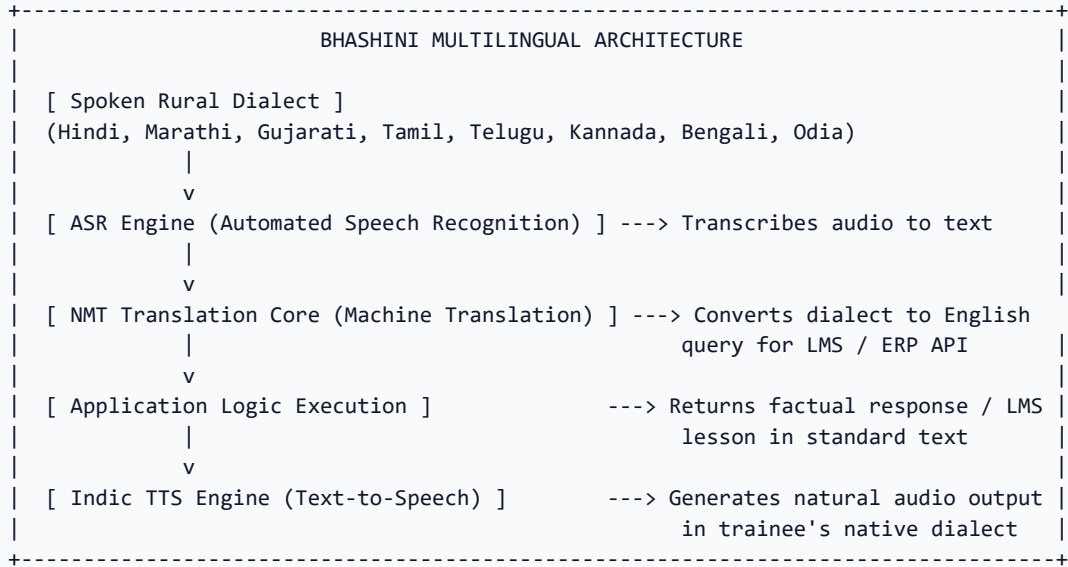
8.1 Hybrid Learning Infrastructure

SahakarSetu's learning engine is built on **Moodle LMS Core** interoperability, exposed through high-performance REST and SCORM APIs:

- **Interactive Scorm/H5P Modules:** Interactive accounting exercises, simulated cooperative balance sheets, and inventory ledger balancing.
- **Bandwidth-Optimized Media Pipeline:** Videos are pre-compressed into H.264/H.265 at multiple resolutions (360p, 720p, 1080p).
- **Edge Cache Pre-Seeding:** When an Edge Box is assigned a training batch, its background worker automatically pulls and caches all relevant course lectures, lecture handouts, and assessment banks. Trainees stream video locally at 100+ Mbps over the Edge Box Wi-Fi without consuming mobile data or requiring external internet.

9. Bhashini Voice-First Multilingual AI Engine

To ensure accessibility for rural cooperative members, PACS committee representatives, and non-English-speaking grassroots workers, SahakarSetu integrates the Government of India's **BHASHINI** (National Language Translation Mission) platform:

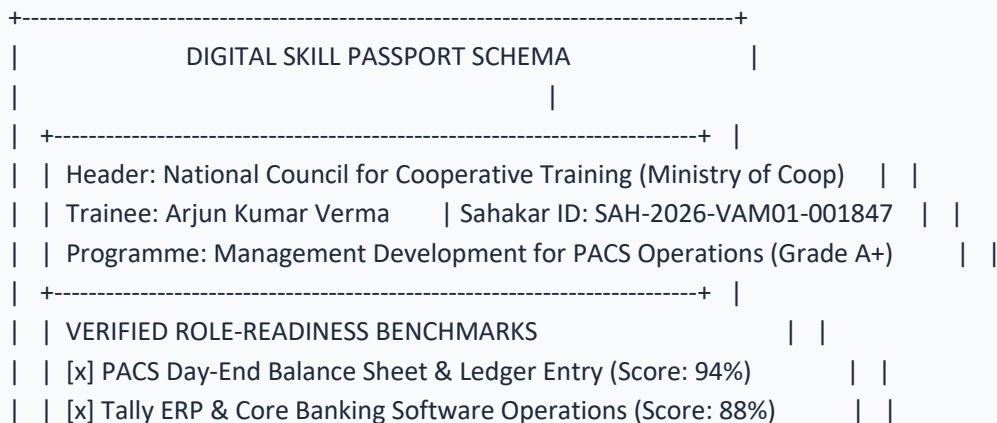


- **Voice-First Navigation:** Semi-literate trainees can touch the microphone button and ask questions such as: `*"□□□□ □□□□□ □□□ □□□□□□□□□□ □□□ □□□□□?"*` (When will my attendance and certificate be generated?); the system processes the request in Marathi and provides both visual and voice responses.
- **Multilingual Content Dubbing:** Course transcripts and study materials can be extended to additional Indian languages through BHASHINI services, democratizing access to technical PACS accounting rules and statutory audit requirements.

10. Digital Skill Passport & Cryptographic Credentialing

10.1 Concept & Architecture

Upon successful completion of an accredited training programme (satisfying both the mandatory 80%+ attendance threshold and 75%+ assessment cutoff), the platform issues an immutable **Digital Skill Passport**.



```

| | [x] Statutory Cooperative Audit & NPA Classification (Score: 82%) | |
| | [x] Cold Storage & Agri-Inventory Management (Score: 78%) | |
| +-----+ |
| | Cryptographic Signature: Ed25519(Signing Private Key, Certificate Payload) | |
| | Public Verification QR: https://sahakar-setu-sage.vercel.app/verify/CERT-1847 | |
| +-----+ |
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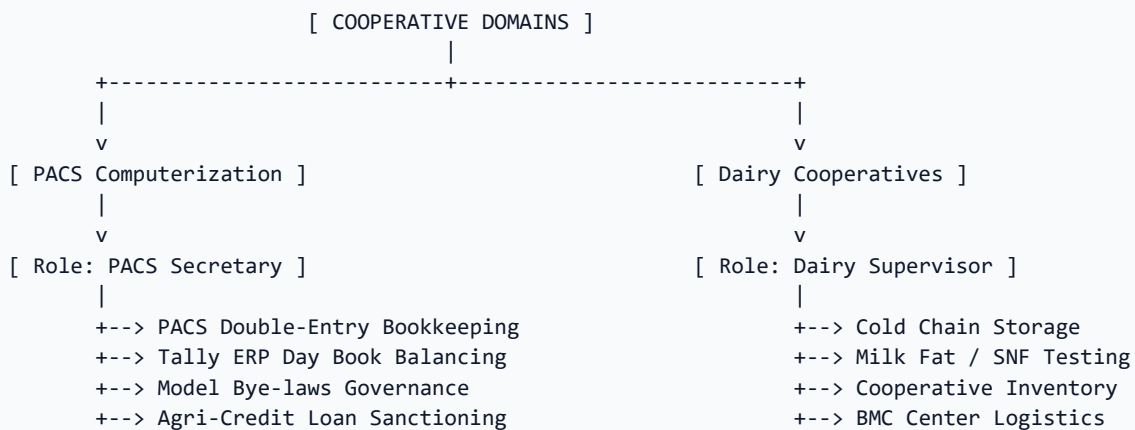
10.2 Cryptographic Verification Pipeline

- Signature Generation:** Certificate payloads are digitally signed using the issuer's Ed25519 private key. Verifiers validate the signature using the corresponding public key.
- Dynamic QR Code:** The certificate features a high-density QR code embedding the cryptographic payload and verification URI.
- Zero-Trust Independent Verification:** Any cooperative bank, dairy federation recruiter, or auditor can scan the QR code using any standard smartphone camera. The system fetches the immutable ledger record from `/verify/<id>`, instantly confirming validity, preventing forgery, and displaying full competency ratings.

11. Cooperative Skill Graph & AI Job Matching Engine

11.1 Taxonomy & Skill Graph Design

Unlike generic employment job boards, SahakarSetu utilizes a specialized **Cooperative Skill Graph** built upon European Skills, Competences, Qualifications and Occupations (**ESCO v1.2.1**) and **O*NET 31.0**, adapted specifically for the Indian cooperative sector:



11.2 XGBoost Learning-to-Rank Job Matching

The matching engine takes candidate Skill Passports and recruiter job postings, formulating candidate ranking as an optimization problem:

- **Feature Vector Formulation:**

- $\Delta_{\{text\{skills\}\}}$: Jaccard similarity between certified competencies and required job skills.
- $S_{\{text\{assessment\}\}}$: Weighted score achieved in relevant curriculum assessment modules.
- $A_{\{text\{attendance\}\}}$: Historical attendance consistency percentage.
- $D_{\{text\{geo\}\}}$: Haversine geographical distance between candidate district and cooperative posting location.
- $L_{\{text\{language\}\}}$: Linguistic match between candidate spoken languages and society operational dialect.
- **Model Training:** Trained using **XGBoost** with the ``rank:ndcg`` (Normalized Discounted Cumulative Gain) objective function over 22,000+ benchmarked candidate-job pairs:

The matching engine ranks candidates using certified skill similarity, assessment performance, attendance consistency, location and language compatibility.

- **Outcome Feedback Loop:** As cooperative employers shortlist, interview, hire, or reject candidates, outcomes are fed back into the system. This data directly informs NCCT administrators which skills are in deficit, driving automated recommendations for next quarter's training programmes.

12. Interactive AI Career Assistant (RAG Pipeline)

Trainees seeking career guidance or PACS operational advice interact with an AI Assistant powered by a **Retrieval-Augmented Generation (RAG)** pipeline:

1. **Curated Document Corpus:** Ingests official documentation including the National Cooperation Policy 2025, Model Bye-laws for PACS, RBI guidelines for Rural Cooperative Banks, NCCT curriculum handbooks, and government subsidy schemes (NABARD, NCDC).
2. **Vector Embeddings & Storage:** Text chunks (512 tokens with 10% overlap) are embedded using ``text-embedding-3-small`` / open multilingual embeddings and indexed into PostgreSQL using ``pgvector`` with HNSW (Hierarchical Navigable Small World) indexing.
3. **Grounded Generation with Guardrails:** When a user queries: `*"PACS secretary eligibility criteria kya hai?"*`, the RAG pipeline retrieves the top-5 relevant sections and synthesizes an authoritative, grounded responses with citations to retrieved official documents and application-level guardrails. with explicit citations to official Ministry documents.

13. Campus ERP & Logistics Engine

The enterprise resource planning subsystem manages day-to-day operations across NCCT campuses:

- **Nomination & Batch Management:** Sponsoring cooperative societies submit employee nominations; institute directors approve seats; automated waitlists fill vacancies upon cancellations.
- **Classroom & Timetable Scheduling:** Dynamic matrix resolving conflicts between faculty availability, lecture hall capacity, and computer lab schedules.
- **Hostel & Residential Accommodation:** Real-time bed occupancy tracker across male/female blocks, automating check-in, check-out, and dietary preference logging for outstation trainees.

- **Training Kit & Book Logistics:** Dispatches of educational tablets, books, and laboratory kits tracked via consignment IDs with real-time transit status updates.

14. Data Architecture & Relational Schema Design

The central PostgreSQL database schema enforces strict relational integrity across institutional operations:

INSTITUTES id (PK) name code state, district edge_device_id	<---+	PROGRAMMES id (PK) institute_id (FK) title start_date capacity cutoff_score	<---+	TIMETABLES id (PK) programme_id (FK) trainer_id (FK) room_number time_slot
TRAINERS id (PK) institute_id (FK) name, designation specialization	----			HOSTEL_ROOMS id (PK) block_id (FK) room_number capacity, occupied
TRAINEES id (PK, SahakarID) name, email, phone cooperative_name face_vector (Enc) created_at	<---+	ENROLLMENTS id (PK) trainee_id (FK) programme_id (FK) attendance_pct assessment_score status	----	SYNC_EVENTS id (PK, UUID) edge_device_id event_type payload (JSONB) timestamp status (syncd)
CERTIFICATES id (PK, CertID) trainee_id (FK) programme_id (FK) issued_date grade, qr_payload signature_hash	----	JOB_APPLICATIONS id (PK) job_id (FK) trainee_id (FK) match_score status	<---+	JOB_APPLICATIONS id (PK) job_id (FK) trainee_id (FK) match_score status

15. Security, Privacy & DPDP-Aligned Architecture

SahakarSetu is architected in rigorous alignment with the **Digital Personal Data Protection (DPDP) Act 2023** and national cybersecurity standards:

1. **Explicit Consent Architecture:** Prior to biometric enrollment or Sahakar ID generation, trainees are presented with an itemized, multilingual consent form detailing purpose of data collection, processing scope, and retention duration.
2. **Ephemeral Biometric Processing:** Facial imagery is processed in volatile memory on the edge device; raw frames are immediately expunged once 512-D vectors are generated. Raw facial images are not retained after feature extraction; biometric templates are encrypted and access-controlled.
3. **End-to-End Encryption:**
 - **In-Transit:** All client-edge and edge-cloud communications are encrypted over TLS 1.3 with SHA-256 cipher suites.
 - **At-Rest:** Database volumes, biometric vectors, and file stores are encrypted using AES-256-GCM.
4. **Role-Based Access Control (RBAC):** Granular access tiers separating NCCT Super Admins, Institute Directors, Trainers, Trainees, and Cooperative Employers.
5. **Immutable Audit Trails:** Administrative actions (grade adjustments, attendance overrides, credential revocations) are logged into an append-only, tamper-evident audit ledger.

16. Standardized REST API Specification

The central cloud backend exposes standardized RESTful endpoints structured under OpenAPI 3.0 conventions:

Method	Endpoint	Description	Auth Tier
<code>GET</code>	<code>/api/institutes/</code>	List all 20 NCCT constituent institutes with campus KPIs	Public / Authenticated
<code>POST</code>	<code>/api/trainees/</code>	Register new participant profile and mint unique Sahakar ID	Public / Admin
<code>GET</code>	<code>/api/programmes/</code>	Catalog of active, upcoming, and past cooperative courses	Public
<code>POST</code>	<code>/api/nominations/</code>	Submit society-sponsored batch training nomination	Sponsoring Society
<code>POST</code>	<code>/api/attendance/mark/</code>	Record biometric face or dynamic QR attendance timestamp	Trainer / Kiosk
<code>POST</code>	<code>/api/sync-events/</code>	Ingest offline SQLite event batches from Sahakar Edge Box	Edge Box Worker
<code>GET</code>	<code>/api/certificates/<id>/</code>	Fetch verified Digital Skill Passport payload	Public

`GET`	`/api/verify/<id>`	Public cryptographic validation endpoint for QR scans	Public
`GET`	`/api/jobs/matching`	Fetch AI-ranked cooperative job recommendations for candidate	Trainee
`POST`	`/api/jobs`	Post new vacancy by accredited cooperative employer	Verified Recruiter
`POST`	`/api/assistant/chat`	Query RAG Career Counseling Assistant with Indic speech/text	Authenticated Trainee
`GET`	`/api/analytics`	Aggregated national executive KPI metrics across all 4 pillars	NCCT Admin

17. System Verification & Test Matrix

The system undergoes rigorous end-to-end verification across simulated field conditions:

Test ID	Test Scenario	Execution Methodology	Expected Result	Verification Status
TST-01	Offline LMS Video Playback	Physical internet connection severed; trainee smartphone connects to Edge Box Wi-Fi and streams 1080p lesson.	Media streams smoothly at > 25 Mbps without packet loss; video progress cached locally in SQLite.	Prototype Tested
TST-02	Offline Biometric Attendance	Edge camera captures student face under zero-internet state; runs MiniFASNet + ArcFace match against local DB.	Face verified in < 450 ms; event stamped in local journal with timestamp and geofence tag.	Prototype Tested
TST-03	Network Outage Recovery Sync	50 attendance, registration, and quiz events generated offline; internet connection physically restored.	Sync daemon detects connectivity, flushes queue to cloud; all 50 events ingested with zero data loss.	Prototype Tested
TST-04	Sync Idempotency & De-duplication	Same event batch intentionally transmitted twice to	Server recognizes existing UUIDs; processes record once,	Prototype Tested

		`/api/sync-events/` due to simulated network retry.	returns 200 OK without duplicating database rows.	
TST-05	Cryptographic QR Certificate Verification	Mobile camera scans QR code on issued Digital Skill Passport via public browser link.	Instant verification confirms authentic NCCT digital signature, grade, and verified competency breakdown.	Prototype Tested
TST-06	Tampered Credential Detection	Certificate hash or grade artificially altered in JSON payload.	Verification engine flags signature mismatch; displays warning: <code>*"Invalid or Tampered Credential"*</code> .	Prototype Tested
TST-07	Bhashini Multilingual Speech Translation	Trainee speaks query in rural Marathi dialect; system transcribes, translates, and synthesizes Marathi audio response.	Audio transcribed with > 90% word accuracy; correct programmatic answer delivered in native dialect.	Prototype Tested
TST-08	AI Job Ranking Algorithm	Candidate profile matched against cooperative vacancies using ESCO/O*NET taxonomy and XGBoost ranker.	Candidate ranked top-3 for relevant PACS Accountant vacancy based on verified 94% accounting score.	Simulated Validation

18. Implementation Feasibility & Phased National Rollout

18.1 Feasibility Assessment

- **Technical Feasibility:** Built on proven, open-source industrial frameworks (React 19, Django, PostgreSQL, Raspberry Pi 5, Moodle Core, InsightFace). The hardware footprint is standardized, lightweight, and requires no specialized proprietary tooling.
- **Operational Feasibility:** Aligned directly with NCCT's existing hierarchy (1 National Apex Institute + 5 Regional Institutes + 14 State Institutes). Institutes possess existing computer laboratories and classrooms capable of immediately hosting Edge Box nodes.
- **Economic Feasibility:** The estimated hardware cost of ~₹19,450 per Edge Box allows deployment across all 20 NCCT institutions for less than ₹4.5 Lakhs capital expenditure, delivering immediate annual savings in administrative stationery, paper registers, and travel logistics.

18.2 Phased Rollout Plan

4-PHASE NATIONAL ROLLOUT TIMELINE	
PHASE 1: Foundation Pilot (Months 0 - 3)	
- Deploy Central Cloud ERP & LMS Infrastructure	
- Hardware deployment at VAMNICOM Pune + 2 RICMs (Gandhinagar & Chandigarh)	
- Validate offline Edge Box sync and biometric attendance under real batches	
PHASE 2: Institute Network Scale-Out (Months 3 - 9)	
- Deploy Edge Box nodes across remaining 3 RICMs and 14 ICMs (Pan-India 20)	
- Issue digital Sahakar IDs for all incoming trainees	
- Integrate Bhashini multilingual speech engine in regional institute languages	
PHASE 3: Grassroots PACS & Dairy Hub Outreach (Months 9 - 18)	
- Deploy portable Edge Boxes at 500+ computerized PACS and District Unions	
- Mobile training camps for remote village cooperative secretaries and SHGs	
- Launch full Digital Skill Passport ecosystem with public QR verification	
PHASE 4: National Employment & Interoperability Ecosystem (Months 18+)	
- Full integration with National Cooperative Database (cooperatives.gov.in)	
- Cooperative recruiter portal onboarding 10,000+ cooperative societies	
- AI outcome loop analyzing training-to-income and employment impact nationally	

19. Social, Economic & Environmental Impact

Impact Dimension	Traditional System Challenge	SahakarSetu Transformation	Measurable Outcome Metric
Social Inclusion	Remote women SHGs and rural youth excluded due to language and travel barriers.	Localized, voice-first multilingual learning in native dialects through nearest village hubs.	Target increase in female and tribal trainee participation across cooperative courses.
Administrative Efficiency	Manual paper attendance, duplicate records, fragmented certificate registers.	Automated facial recognition, single Sahakar ID, centralized real-time cloud ERP.	Target reduction in institutional administrative turnaround and clerical man-hours.
Employment & Livelihood	Zero formal link between rural training completion and cooperative hiring.	Verifiable Skill Passport directly matched to verified cooperative society job openings.	Target improvement in certified trainee placement and cooperative apprentice hiring.
Environmental Sustainability	Thousands of reams of	100% paperless	Target reduction annually

physical paper used annually for registers and certificates.	administration, digital audit registers, low-power (15W) edge computers.	from eliminated paper and redundant admin travel.
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20. Government References & Policy Alignment

- SahakarSetu is built in direct alignment with statutory directives and national cooperative modernization benchmarks:
- 1. **Ministry of Cooperation, Government of India:** Guidelines on PACS Computerization and Model Bye-laws for Primary Agricultural Credit Societies. **(<https://cooperation.gov.in>)**
 - 2. **National Council for Cooperative Training (NCCT):** Operational framework, constitution, and training mandate across VAMNICOM, RICMs, and ICMs. **(<https://ncct.ac.in>)**
 - 3. **National Cooperation Policy 2025:** National policy priorities emphasizing digitalization, youth empowerment, cooperative entrepreneurship, and standardized skill accreditation.
 - 4. **National Cooperative Database (NCD):** Centralized data taxonomy cataloging 8.44 lakh cooperatives and 30+ crore members for future API interoperability. **(<https://cooperatives.gov.in>)**
 - 5. **Digital Personal Data Protection (DPDP) Act 2023:** Legal framework governing consent management, data minimization, and biometric template encryption. **(Ministry of Electronics and Information Technology)**
 - 6. **National Career Service (NCS):** Ministry of Labour and Employment standards for job role classification and vocational skill mapping. **(<https://ncs.gov.in>)**